Deployment Operations Runbook—Enterprise GenAI Platform

Version 2026.1 | Azure DevOps / GitHub Actions | LC 3 Project Module

Classification: INTERNAL USE | On-Call Reference | Last Updated: July 2026

Runbook Overview

This runbook provides standardized, step-by-step procedures for deployment operations and incident response across the National Retail Corp GenAI platform. All procedures assume Azure-native tooling (Azure DevOps, Azure Monitor, Application Insights, Azure API Management, AKS) and align with Microsoft SRE practices and the Azure Well-Architected Framework.

Emergency Contacts (On-Call Rotation):
• **SRE On-Call:** sre-oncall@nrc.internal | PagerDuty: NRC-GenAI-SRE
• **VP Platform Engineering:** platform-vp@nrc.internal (escalation after 15 min for P0/P1)
• **CISO Security Operations:** soc@nrc.internal (security incidents only)
• **App Owners:** askretail-owner@nrc.internal | stocksense-owner@nrc.internal | hrhelper-owner@nrc.internal

1. Standard Deployment Procedure

Use this procedure for scheduled releases of models, prompts, orchestration logic, or infrastructure changes.

1.1 Pre-Deployment Checklist

Step	Action	Verification
1.1.1	Confirm release version in Azure DevOps / GitHub Release page. Verify semantic version (for example, v2.3.1-askretail).	Release page shows green build; artifact bundle SHA matches commit.
1.1.2	Review change log and risk assessment. Confirm no "latest" tags in model references.	Model reference JSON shows pinned version (for example, gpt-4o-2026-01-15).
1.1.3	Verify evaluation gate report is attached to release.	Evaluation report: groundedness,

	All automated gates must show PASS.	hallucination, latency, security scan = PASS.
1.1.4	Check maintenance window. Tier 1 apps: deploy 02:00–04:00 ET Tue/Thu only. Tier 2: deploy 06:00–08:00 ET Mon/Wed.	Current time falls within approved window; no conflicting change requests.
1.1.5	Verify backup of current production state: container image digest, prompt bundle version, model reference, API Management configuration.	Backup manifest stored in Azure Blob 'production-backups/{app}/{date}'.
1.1.6	Notify stakeholders via #genai-deployments Slack channel. Include app name, version, change summary, rollback ETA.	Slack message posted; @channel for Tier 1, @app-owner for Tier 2.
1.1.7	Confirm on-call SRE is available and aware of deployment. For Tier 1-S (HRHelper), confirm Compliance Officer standby.	On-call acknowledged in PagerDuty; Compliance Officer confirmed (HRHelper only).

1.2 Deployment Execution

Step	Action	Verification
1.2.1	Trigger deployment pipeline via Azure DevOps Release / GitHub Actions workflow_dispatch. Select target environment: Staging or Production.	Pipeline run initiated; run ID captured in deployment log.
1.2.2	Monitor staging deployment (if applicable). Run smoke tests: 5 synthetic requests per major endpoint.	Smoke test report: 5/5 PASS; latency within baseline ±10%.
1.2.3	Obtain production approval (if required per Approval Matrix). For Tier 1: VP Platform Eng + App Owner. For Tier 1-S: + AI Ethics + Legal.	Approval recorded in Azure DevOps Environment Approvals; timestamp logged.
1.2.4	Initiate progressive rollout per app strategy (see Section 2). Do NOT skip monitoring windows.	Traffic split active; monitoring dashboards open in Grafana / Azure Monitor.

1.2.5	Execute post-deploy validation (Section 1.3).	All health checks PASS.
1.2.6	If validation passes, complete traffic cutover. Update Azure App Configuration production flag. Announce completion in Slack.	100% traffic on new version; previous version kept warm for 30 min (blue/green) or 2 hours (canary buffer).
1.2.7	Update CMDB and deployment registry. Close change request in ServiceNow.	CMDB reflects new version; CR status = Closed Complete.

1.3 Post-Deploy Validation Checklist

Check	Method	Pass Criteria
Synthetic Health	Application Insights Availability Test (3 probes: US East, West 2, Central)	100% pass rate; p50 latency < SLA
Error Rate	Azure Monitor Metrics: requests/failed (5-min window)	< 0.1% for Tier 1; < 1% for Tier 2
Groundedness Baseline	Azure AI Evaluation continuous runner (first 50 samples)	Score ≥ threshold - 0.02 (tolerance)
Token Consumption	API Management + Azure Cost Management	Cost per request < 110% of baseline
Pod Health	AKS Container Insights: running pods vs. desired	100% match; 0 CrashLoopBackOff
API Gateway Routing	APIM Test Console: 3 live requests per app	200 OK; correct backend routed; headers valid
Security Posture	Defender for Cloud: vulnerability scan of new image	0 Critical, 0 High findings

2. Progressive Rollout Checklist

Each app uses a differentiated rollout strategy based on risk tier. This section provides per-strategy checklists.

2.1 AskRetail—Blue/Green Rollout

Phase	Action	Duration / Criteria
Pre-Cutover	Verify green environment (new version) is fully provisioned. Run full synthetic test suite (20	15 min

	scenarios). Confirm blue environment (current) is stable baseline.	
Traffic Switch	Update API Management backend policy: route 100% traffic to green. Monitor for 5 minutes.	5 min
Observation Window	Watch: error rate, latency p99, groundedness score, hallucination rate, token cost. Compare against blue baseline.	15 min
Validation Gate	All metrics within ±10% of baseline? If YES → proceed to decommission blue. If NO → immediate rollback (Section 4).	5 min
Decommission Blue	Scale blue environment to 0 replicas. Retain artifact for 7 days (rollback safety buffer). Update DNS / load balancer.	10 min
Post-Rollout	Monitor for 1 hour. Announce completion. Update runbook with any anomalies.	1 hour

2.2 StockSense—Canary Rollout

Phase	Action	Duration / Criteria
10% Canary	Route 10% traffic (ad-hoc queries + 1 batch job) to canary pods. Monitor: error rate, batch completion time, MAPE on canary subset.	30 min
Canary Gate	Error rate < 1%? Batch completed within 45 min? MAPE within ±1% of baseline? If YES → promote to 50%. If NO → rollback.	10 min
50% Expansion	Route 50% traffic. Run 3 full batch jobs (morning, midday, evening). Monitor forecast accuracy across all SKUs.	2 hours
50% Gate	All batch jobs complete on time? MAPE < 8%? No integration test failures? If	15 min

	YES → promote to 100%. If NO → rollback to 10% or 0%.	
100% Cutover	Route 100% traffic. Monitor for 1 hour. Keep previous version warm for 2 hours.	1 hour + 2 hour buffer
Decommission	Scale previous version to 0. Archive batch outputs for 90 days (SOX compliance).	15 min

2.3 HRHelper—Blue/Green with Extended Observation

Phase	Action	Duration / Criteria
Pre-Cutover	Green environment verified. Content safety filters validated. Audit trail logging confirmed. Legal Counsel notified.	20 min
Traffic Switch	Route 100% to green. Monitor for 30 minutes (extended due to sensitivity).	30 min
Observation Window	Watch: error rate, content safety violations (must be 0), PII leakage (must be 0), escalation accuracy, latency.	30 min
Validation Gate	Zero safety violations? Zero PII leaks? Escalation accuracy ≥ 95%? If YES → decommission blue. If NO → immediate rollback + notify Legal.	10 min
Decommission Blue	Scale blue to 0. Preserve all conversation logs (immutable WORM storage). Update compliance registry.	10 min
Post-Rollout	Monitor for 2 hours. Compliance Officer sign-off required before closing deployment ticket.	2 hours

3. Incident Response Procedure

Follow this procedure for all production incidents affecting GenAI platform availability, quality, or security.

3.1 Incident Severity Classification

Severity	Criteria	Response Time	On-Call Action
SEV-0 (Critical)	Complete outage of Tier 1 app; PII breach; security incident	< 5 min	Page immediately; war room within 10 min
SEV-1 (High)	Severe degradation (>50% error rate or >3× latency SLA); safety violation	< 15 min	Page immediately; war room within 30 min
SEV-2 (Medium)	Partial degradation; quality regression (groundedness drop > 0.05); cost anomaly	< 1 hour	Slack alert + ticket; war room optional
SEV-3 (Low)	Minor threshold breach; single pod failure; non-impacting alert	< 4 hours	Ticket for next business day

3.2 Incident Response Steps (SEV-0 / SEV-1)

Step	Action	Owner
1. Detect	Alert fires (Azure Monitor / PagerDuty). On-call SRE acknowledges within 5 min. Create incident channel #incident-{timestamp}.	On-Call SRE
2. Triage	Classify severity using table above. Notify stakeholders: App Owner (all), VP Platform (SEV-0/1), CISO (security), Legal (HRHelper SEV-0/1).	On-Call SRE
3. Contain	If SEV-0/1: execute immediate rollback (Section 4) OR enable circuit breaker (API Management rate limit to 0 for affected app). Preserve logs/evidence.	On-Call SRE + Platform Eng
4. Communicate	Post initial status to status.nrc.internal within 10 min. Update every 30 min until resolved. For SEV-0, notify executive team within 15 min.	Incident Commander
5. Investigate	Use traces (Application Insights), logs (Log	Platform Eng + App Owner

	Analytics), and dashboards (Grafana) to isolate root cause (Section 5). Do NOT modify production state without approval.	
6. Resolve	Apply fix or complete rollback. Verify health via Section 1.3 checklist. Monitor for 1 hour (SEV-1) or 4 hours (SEV-0) before all-clear.	Platform Eng
7. Close	Update status page. Send closure notification. Schedule post-mortem within 24 hours (SEV-0) or 48 hours (SEV-1).	Incident Commander

4. Rollback Execution Procedure

Rollback is the primary containment action for deployment-related incidents. Execute without hesitation if health checks fail during rollout.

4.1 Automated Rollback Triggers

- The following conditions trigger automatic rollback via Azure DevOps Gates or GitHub Actions automated checks:
- Post-deploy health check failure: Synthetic tests < 100% pass for > 3 consecutive probes.
- Error rate spike: > 5× baseline for > 5 minutes.
- Latency breach: p99 > 1.5× SLA for > 5 minutes (Tier 1) or > 10 minutes (Tier 2).
- Hallucination rate > 3% in canary window.
- Content safety violation severity ≥ 4 (HRHelper).

4.2 Manual Rollback Steps

Step	Action	Verification
4.2.1	Identify last known good version from deployment registry. Confirm artifact exists in Azure Container Registry and artifact store.	Image digest and prompt bundle SHA verified.
4.2.2	For Blue/Green: switch API Management traffic 100% to blue (previous) environment. For Canary: set canary weight to 0%.	APIM policy updated; traffic routing confirmed via Application Insights.
4.2.3	Revert Azure App Configuration to previous version flags (model	Config read by pods within 30 seconds; no 404 errors.

	reference, prompt version, feature flags).	
4.2.4	If artifact revert required: re-deploy previous container image to green/canary environment. Verify pod health.	kubectl get pods shows Running; readiness probe PASS.
4.2.5	Run smoke tests (5 synthetic requests per endpoint).	5/5 PASS; latency within baseline.
4.2.6	Monitor for 15 minutes. Confirm error rate, latency, and quality metrics return to baseline.	Dashboards show green status; no new alerts.
4.2.7	Notify stakeholders: "Rollback to vX.Y.Z complete. System stable." Update incident ticket.	Slack posted; ticket updated; status page green.
4.2.8	Preserve failed deployment artifacts and logs for post-mortem. Do NOT delete failed release for 30 days.	Artifacts in 'failed-deploys/{app}/{date}' container.

5. Root Cause Investigation Guide

Use this decision tree to isolate whether an incident stems from model changes, retrieval failures, orchestration bugs, or infrastructure issues.

5.1 Investigation Decision Tree

```
[Incident Detected]
    |
    ├── Is error rate elevated across ALL apps?
    |   ├── YES → Infrastructure issue (AKS, API Management,
network)
    |   |   └── Check: AKS node health, API Management
capacity, VNet connectivity
    |   └── NO → App-specific issue
    |       |
    |       ├── Is latency elevated but error rate normal?
    |       |   ├── YES → Model or retrieval bottleneck
    |       |   |   └── Check: Azure OpenAI token rate limits,
AI Search latency, embedding model queue depth
    |       |   └── NO →
    |       |       |
```

```
    |       |       ├── Are quality metrics degraded
(groundedness, hallucination)?
    |       |       |   ├── YES → Model or prompt regression
    |       |       |   |   └── Check: Model version change,
prompt template diff, retrieval index version
    |       |       |   └── NO → Orchestration or data issue
    |       |       |       └── Check: Orchestrator logs,
downstream API failures (SAP, Synapse), prompt injection
attempts
    |       |       |
    |       |       └── Is there a spike in token consumption
or cost?
    |       |           ├── YES → Inefficient prompt or model
change
    |       |           |   └── Check: Prompt token count
trend, model swap (e.g., GPT-4o → o3-mini)
    |       |           └── NO → Continue deep trace analysis
    |       |
    |       └── Are there security alerts (DLP, content
safety)?
    |           ├── YES → Security incident; involve CISO
    |           |   └── Check: Sentinel alerts, APIM access
logs, Entra ID sign-in anomalies
    |           └── NO → Continue monitoring; may be transient
```

5.2 Key KQL Queries for Investigation

Query 1: Error rate by app and version (last 1 hour)

```
requests
| where timestamp > ago(1h)
| where cloud_RoleName in ("askretail", "stocksense",
"hrhelper")
| summarize errorRate = countif(success == false) * 100.0 /
count() by cloud_RoleName, application_Version, bin(timestamp,
5m)
| render timechart
```

Query 2: Latency distribution by model version

```
customEvents
| where name == "ModelInference"
| where timestamp > ago(1h)
| extend modelVersion =
```

```
tostring(customDimensions["ai.model.name"])
| summarize latency_p50 = percentile(duration, 50), latency_p99
= percentile(duration, 99) by modelVersion, bin(timestamp, 5m)
| render timechart
```

Query 3: Groundedness score trend

```
customMetrics
| where name == "ai.groundedness.score"
| where timestamp > ago(24h)
| summarize avgGroundedness = avg(value) by cloud_RoleName,
bin(timestamp, 1h)
| render timechart
```

Query 4: API Management throttling and quota events

```
AzureDiagnostics
| where ResourceProvider == "MICROSOFT.APIMANAGEMENT"
| where OperationName == "GatewayLog"
| extend responseCode = toint(responseCode_d)
| where responseCode == 429 or responseCode == 503
| summarize count() by apim_name_s, backendURL_s,
bin(TimeGenerated, 5m)
```

6. Post-Incident Verification and Post-Mortem

6.1 Post-Incident Verification Checklist

Check	Method	Owner
System Health	Run full Section 1.3 post-deploy checklist. All checks must PASS.	On-Call SRE
Metric Baselines	Confirm all metrics (latency, error rate, quality, cost) within normal range for 24 hours post-incident.	Platform Eng
Log Integrity	Verify no log gaps during incident window. Ensure audit trails are complete (especially HRHelper).	Platform Eng

Cost Impact	Document any overage costs from incident. Update Cost Management alerts if thresholds were breached.	FinOps Analyst
Stakeholder Notification	Send all-clear to App Owners, executives, and customers (if external impact).	Incident Commander
Artifact Preservation	Lock incident-related logs, traces, and artifacts in WORM storage for compliance.	Platform Eng

6.2 Post-Mortem Template

Schedule: SEV-0 within 24 hours; SEV-1 within 48 hours; SEV-2 within 5 business days.

Section	Content Guidance
Incident Summary	What happened? One-paragraph overview. Include app, version, time range, and customer impact.
Timeline	Minute-by-minute from detection to resolution. Include alert firing, war room start, rollback decision, all-clear.
Root Cause	Technical root cause (not proximate cause). Use 5 Whys. Categorize: Model, Retrieval, Orchestration, Infrastructure, Security, Human Error.
Impact Assessment	Downtime minutes, error count, affected users, revenue impact, data exposure (if any).
Detection	How was the incident detected? Was alert timely? Any gap in monitoring?
Response Effectiveness	Did rollback work? Were runbook steps clear? Any improvisation required?
Lessons Learned	What went well? What could improve? Be specific, not generic.
Action Items	Numbered list with owner and due date. Tag JIRA tickets. Categories: Prevent, Detect, Respond, Mitigate.
Blamelessness Statement	Explicit statement that post-mortem is blameless. Focus on system improvements, not individual fault.

7. Quick Reference—One-Page Cheat Sheet

App	Rollout Strategy	Rollback Time (Target)	Key Monitoring URL
AskRetail	Blue/Green	< 3 minutes	https://grafana.nrc.internal/d/askretail-prod
StockSense	Canary (10% → 50% → 100%)	< 5 minutes	https://grafana.nrc.internal/d/stocksense-prod
HRHelper	Blue/Green (extended 30 min)	< 3 minutes	https://grafana.nrc.internal/d/hrhelper-prod

Runbook Maintenance: This runbook is reviewed monthly by the Platform SRE team and updated after every SEV-0/1 incident. Submit change requests via #genai-runbook-changes Slack channel.

References: Microsoft Azure SRE Playbook 2026, Azure Well-Architected Framework—Operational Excellence, Google SRE Book (Chapter 14: Incident Management), NIST SP 800-61 Rev. 2 (Computer Security Incident Handling Guide).